

Curriculum Vita

RESEARCH STATEMENT

I research **software engineering** with an emphasis on advancing **software security**. In particular, I study the role, influence, and use of large language models in ensuring secure software. Previously, I have also explored the human factors impacting engineering practices.

Research Interests: AI for Software Engineering, Software Security, Vulnerability Detection and Repair, Natural Language Processing, Large Language Models, Data analysis, Human Aspects of Software Engineering

EDUCATION

Drexel University Ph.D. in Computer Science	Pennsylvania, USA 2022–Present
Azad University Karaj Branch B.Sc. in Computer Engineering	Alborz, IRN 2018–2021
Azad University Qazvin Branch B.Sc. in Mechanical Engineering	Qazvin, IRN 2016–2018

SELECT COURSEWORK

– Intro to Artificial Intelligence	– Software Analytics
– Machine Learning	– Cloud Software Engineering
– Software Design	– Fundamentals of Safe Computing

RESEARCH EXPERIENCE

Research Assistant SOAR Lab - Drexel University 	Pennsylvania, USA September 2022–Present
– Developing LLM-based agentic frameworks for proof-of-concept vulnerability detection and exploitation. – Developing LLM-enhanced taint analysis solutions for automated vulnerability detection. – Investigated the security of LLM-generated software patches, identifying the LLM-specific vulnerability patterns. – Empirically evaluated the security awareness of multiple LLMs when responding to programming questions. – Conducted a systematic mapping study on the use of psycholinguistic tools like LIWC in software engineering. – Conducted an empirical study on interpersonal trust among open-source developers across the GitHub platform – Qualitatively analyzed the textual communications of developers and published sentiment analysis dataset	
Quantitative Development and Research Intern K2-Quant LLC 	Arizona, USA July 2026–September 2026
– Developed and analyzed quantitative trading strategies. – Built research pipelines for financial data analysis, backtesting, and strategy evaluation. – Developed software tools for portfolio analysis, market research, and investment decision support. – Analyzed historical performance, risk, and robustness across varying market conditions.	

PUBLICATIONS

- **Amirali Sajadi**, T. Nguyen, K. Damevski, and P. Chatterjee, “**AXE: Grey-Box Exploitability Confirmation for Localized Vulnerability Reports**,” *Under Review*.
- **Amirali Sajadi**, S. Dutta, and P. Chatterjee, “**An Empirical Analysis of CodeQL False Positives and Query Refinements for Java Vulnerabilities**,” *Under Review*.
- **Amirali Sajadi**, T. Nguyen, K. Huynh, E. Parra, and P. Chatterjee, “**TraceView: Interactive Visualization of Agentic Program Repair Trajectories**,” Accepted to the IEEE International Conference on Software Maintenance and Evolution (ICSME), 2026.
- **Amirali Sajadi**, K. Damevski, and P. Chatterjee, “**Are AI-Generated Fixes Secure? Analyzing LLM and Agent Patches on SWE-bench**,” *Under Review*
- **Amirali Sajadi**, B. Le, A. Nguyen, K. Damevski, and P. Chatterjee, “**Do LLMs Consider Security? Empirical Study on Responses to Programming Questions**,” *Accepted to Empirical Software Engineering (EMSE)*, Springer, 2025
- **Amirali Sajadi**, K. Damevski, and P. Chatterjee, “**Psycholinguistic Analyses in Software Engineering Text: a Systematic Mapping Study**,” *Accepted to Information and Software Technology (IST)*, 2025
- **Amirali Sajadi**, K. Damevski, and P. Chatterjee, “**Interpersonal Trust in OSS: Exploring Dimensions of Trust in GitHub Pull Requests**,” Proceedings of the 45th International Conference on Software Engineering (ICSE), New Ideas and Emerging Results Track, May 2023 (Acceptance rate: 22%)
- **Amirali Sajadi**, K. Damevski, and P. Chatterjee, “**Towards Understanding Emotions in Informal Developer Interactions: A Gitter Chat Study**,” The 31st ACM Joint European Software Engineering Conference and Symposium on the Foundations of Software Engineering (ESEC/FSE), Ideas, Visions and Reflections Track, Dec 2023

RESEARCH PRESENTATION

- **Interpersonal Trust in OSS: Exploring Dimensions of Trust in GitHub Pull Requests**, 45th International Conference on Software Engineering (ICSE), Paper Presentation Talk, via Zoom.
- **Towards Understanding Emotions in Informal Developer Interactions: A Gitter Chat Study**, 31st ACM Joint European Software Engineering Conference and Symposium on the Foundations of Software Engineering (ESEC/FSE), Paper Presentation Talk, California, USA.

TEACHING EXPERIENCE

Under the supervision and in collaboration with my advisor, served as graduate research mentor for four undergraduate students. Worked closely with the students to perform qualitative and quantitative analyses.

- **John Nguyen**, Contributed to research and tool development for vulnerability exploitability analysis, 2026
- **Kimmie Huynh**, Contributed to the development and evaluation of an interactive visualization tool for agentic program repair trajectories, 2026
- **Moon Mustafa**, Performed qualitative analyses on open-source issue resolution processes, 2025
- **Binh Lee, Anh Nguyen**, Performed data analysis and a case study on the security of the LLM-generated content, 2024
- **Vanessa Martinez**, Curated a dataset for sentiment analysis, using developer chat utterances, 2023

Additionally, since 2024, I have been a course instructor and a teaching assistant, supporting students in their learning and development in computer science courses.

SERVICE TO PROFESSION

- **Reviewer**
 - IEEE Transactions on Software Engineering (TSE), 2026
 - ACM Transactions on Software Engineering and Methodology (TOSEM), 2026
 - Software Testing, Verification and Reliability (STVR), 2025
 - IEEE Software, 2024
- **Review Committee Member**
 - Artifact Evaluation Committee, 46th International Conference on Software Engineering (ICSE), 2024

FELLOWSHIPS AND AWARDS

- **Drexel Graduate Leadership Award, 2025**, in recognition of outstanding achievements during the 2024-2025 academic year.
- **ACM -SIGSOFT CAPS Travel Award** ACM Special Interest Group on Software Engineering to financial support for selected researchers in attending the 31st ACM Joint European Software Engineering Conference and Symposium on the Foundations of Software Engineering (ESEC/FSE)
- **Drexel CCI Travel Award, 2022**, Department funding - covering 50% of conference travel costs for ICSE 2023
- **GSA Graduate Travel Subsidy Award, 2023**, University funding - covering 50% of conference travel costs for ESEC/FSE 2023

PROFESSIONAL MEMBERSHIP & AFFILIATION

- **Member, Doctoral Student Association (DSA), Drexel University**
Executive Board Member and Events Coordinator at Doctoral Student Association (DSA) Drexel University College of Computing and Informatics, 2023-2024
- **Member, Association for Computing Machinery (ACM), 2022-Current**
- **Member, Upsilon Pi Epsilon (UPE), Drexel Chapter, 2024-Current**

TECHNICAL SKILLS

Languages: Python, Java, JavaScript, Go, SQL, C, Bash

Tools: Git, Docker, PostgreSQL, MySQL, SQLite, CodeQL, Semgrep, Bandit, Quantconnect Lean

Libraries: LangGraph, Pandas, NumPy, Matplotlib, PyTorch, Keras, Flask, jQuery, Bootstrap, Android